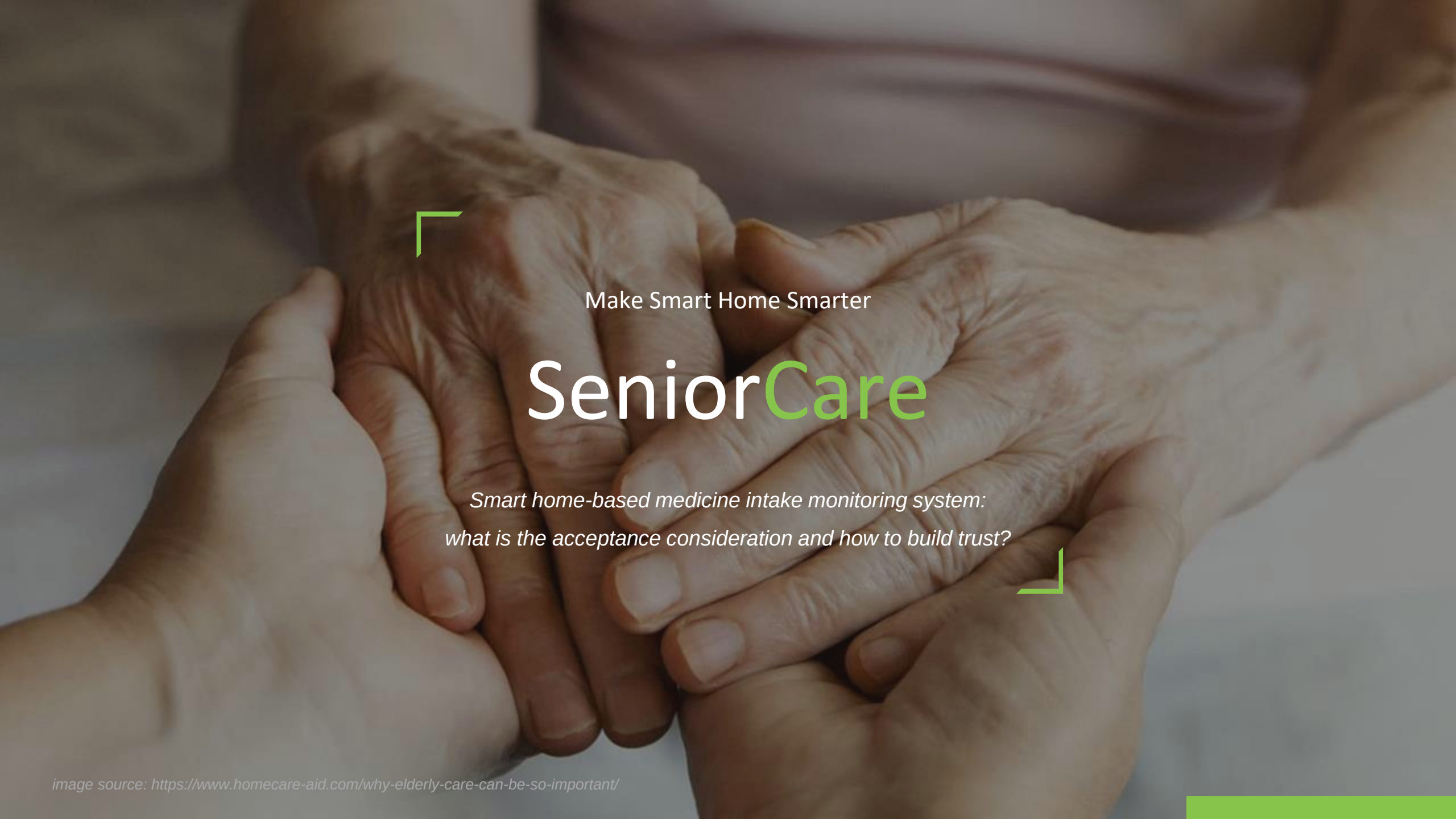




Make Smart Home Smarter

SeniorCare

*Smart home-based medicine intake monitoring system:
what is the acceptance consideration and how to build trust?*

Literature Review.

Smart Home can be regarded as a Service (SHaaS) in the application of the Internet of Things (IoT), which improves security, efficiency, economy, convenience, and comfort within the house and the management of life[1].

Smart home for aged care

developed to help the aged group live independently during Covid 19 and improve their quality of life. The study indicated several “smart home” solutions for the diagnosis and prevention of falls in the aged population[2]

Healthy diet

Through a variety of researches on food image identification approaches, a Linear Distance Coding (LDC) framework is established to help improve low-pixel image recognition performance[3].

Healthcare monitor

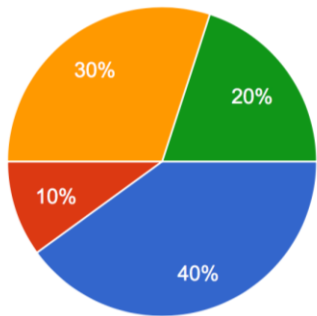
Smart healthcare systems shift hospital treatment to household care, monitoring people’s daily activities on the foundation of analytics algorithms. These health data will provide better family support for caretakers[5].

Pills intake monitor

Research shows that smart home technologies contain a variety of applications. Food and medicine intake monitoring has challenges in activity monitoring and data upload[4].

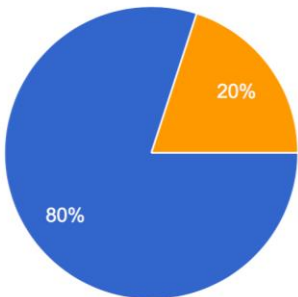
Previous Survey.

How do you recognise the bad food or unhealthy food?
(10 条回复)



- Following my own experience by observing its appearance and smell
- Ask professional consultant or doctor
- Reading ingredients list and best before date
- Search google or other online platforms

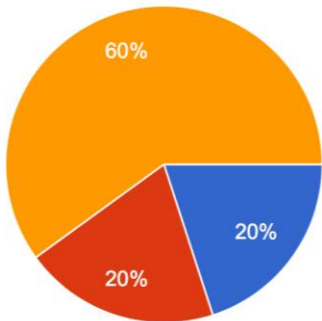
Would you like to try online food care system?
(10 条回复)



- Yes. It is a good way to keep food security and healthy
- No. I prefer traditional approaches to storage food
- Maybe. I have no idea

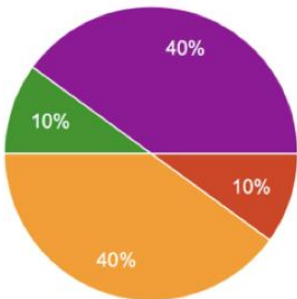
Related to food intake monitor

How do you keep on healthy diet?
(10 条回复)



- Follow my doctor's advice
- Follow my own diet habit (vegetarian or meatatarian)
- Follow online food menu or online professional recommendation

Who do you think are likely to have a healthy diet?
(10 条回复)



- people with medical conditions with specific dietary needs
- Pregnant women
- Baby and children
- Senior or disabled groups
- Everyone

Previous Interview.

Related to food intake monitor

“1. What do you think a smart home looks like currently and what tends to be in the future?”

As far as I am concerned, smart home is related to the self-controlled devices by using some high technologies like AI, cloud computing, machine learning and so on. I think smart home will be much safer due to the advanced smart camera, face recognition, voice control and many different sensors such as smoke sensor and temperature sensor. And in the near future, I wish to open the door just say “Ali Baba, Open, Sesame” without any key.

“2. Have you tried some smart home technologies? If so, what are they?”

I have tried the voice-controlled (smart speaker) and it can play music for me. I really enjoy music when I am cooking, but a voice-controlled devices would be better to use rather than a remote-controlled rice cooker, refrigerator or oven.

“3. Why or why not using this kind of smart home technology?”

There are so many apps to control different smart devices that made me feel annoying when using smart devices. I will never use them in a bunch of applications. It is so ridiculous. Look at my mobile phone. First is for air-conditioner, second is for television, third is for fridge, fourth is for laundry. Some are using Bluetooth, while others link to the home WIFI. If there is no internet in my household, I can’t use them anymore. It is really inconvenient. I want to use all in one app that I can control my electrical appliances in one device or system without connecting WIFI.

“1. What do you think a smart home looks like currently and what tends to be in the future?”

Well, in my view, smart home is like something the science fiction film. All devices are automatically, like cleaning robot, auto cooker, voice controlled refrigerator etc.

“2. How about smart healthcare monitor? Have you tried any smart healthcare technologies? If so, what are they?”

I think healthcare monitor is like some devices that can monitor my physical and mental health, including blood pressure, heart rate, blood oxygen, nutrition intake, dietary, sleep quality, etc. I have used the healthcare app in mobile phone that can track my daily walking activities and track the sleep status (heart rate and breath).

“3. Why using this kind of smart healthcare technology and why do you think your senior doesn’t like to use them?”

For me, I like to use these apps due to their convenience and efficiency and they can give me some advice of daily healthcare. But for my grandma, I think it may be difficult for her to learn and understand how to use it. She hardly uses smart phone or other smart technologies in daily life. As senior group, my grandma told me, “I use mobile phone but not very frequently, just for calling my daughter. Although it takes much time to learn to use apps on mobile phone, I am still willing to learn new technology.” But for her growing age and poor eyesight, she may be hard to read too much words clearly on the mobile phone screen, because she always asked me to help her read the message on the mobile phone.

Previous Interview.

Related to food intake monitor

“4. Do you have senior friends or relatives around you?”

Yes, my grandma, 79 years old.

“5. How diets have changed when she retired?”

She loves all types of food, especially meat. I think the types of food are as usual, not changed greatly. But she eats quite a lot, almost twice as much as before. I care about her overweight. And she loves to eat with families. Sometimes she asked me to accompany her and share dinner. I think during the Covid situation, she wants to get more supports from the family.

“6. Why do you think family support is important during Covid situation for senior? And what support should be provided as a family member?”

I think family support is very important for me during Covid situation. I know my grandma was living alone last year that it is really a tough journey. She needs to control diet and take pills everyday. But no one could help her or remind her. I think if I lived with her, I can remind her of health diet and taking pills on time.

“7. What types of food do you think are good for senior?”

Fresh fruits, fresh vegetables, beef, pork and milk, any nutrition food.

“8. What food should senior avoid? Why?”

Food with a lot of salt and oil may cause physical disorders.

“9. How do you get the information on foods good or bad for senior?”

Online, website. When I have questions, I always search on Google. But I am not sure how accurate the information is. Maybe when I am old, I would like to follow the doctor's advice.

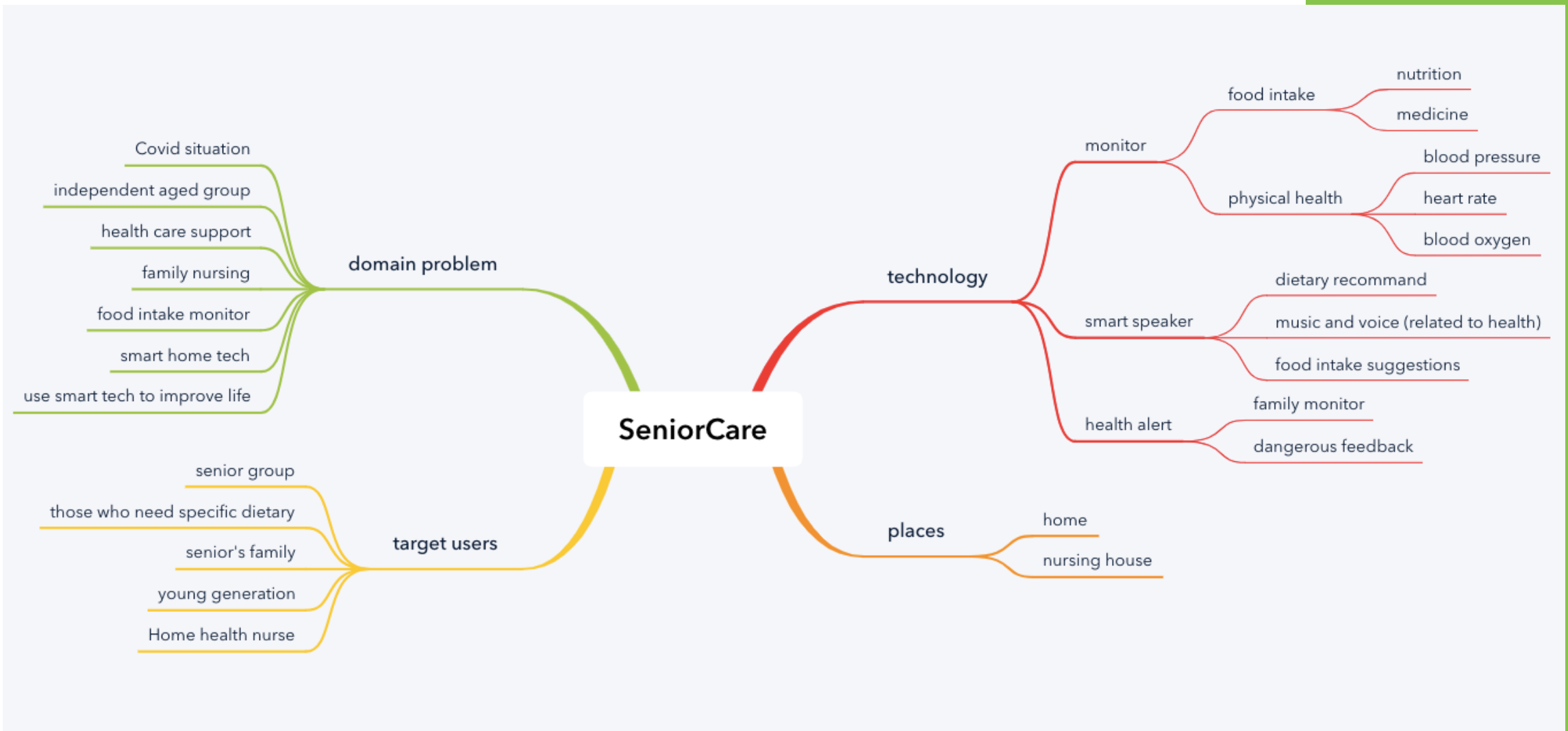
“10. Why do you get information in this way?”

Emmm... I think professional advice with high authority will keep me and my grandma healthy. They are trustful, credible, dependable, fair and scientific. Some information online maybe the advertisement or promotion that persuade to buy useless things. So, I don't trust online information if it is not proved.

“11. If there is a technology to help senior detecting food to ensure a healthy diet, would you like to try? Why?”

I won't try any technology unless it is approved by the government or the authorities. I am worried about its accuracy of datasets and information. It would be dangerous if it is broken and provides some fake news. I would like to try new technology if I am young, because I like to reach new things and to learn latest promising technologies. But I will never let my grandma involved in any risks under the unknown technology without any test or evaluation.

Mindmap.



5+
responses



Smart home-based technology for older adults can play a vital role in family healthcare. Devices that monitor medicine intake and their health have become increasingly important after COVID-19, especially for independent older adults.

To sum up.

Experience of senior using smart home-based technology

Family **support** is important

Medical care plays a crucial role in family support, contributing to healthy aging and a happy family.



Smart home
smart technologies everywhere



Introduction.

IoT, AI, Cloud Computing,
Machine Learning

the smart home industry, especially the home appliance industry, has been leading to technological innovation in the past few years.

Smart home based Healthcare include:

nutrition-aware cooking, special dietary monitor, pills and food intake monitor, eco-friendly system control, and health care monitoring (heart rate, blood pressure, etc)



Special dietary for
senior group

Upload the daily medication schedule
and doctor's instructions online,
identify the medication intake periods,
and set an alarm as a reminder.

SeniorCare System.



It's a plastic
box with



Speaker



Microphone



Detection sensors

Smart Speaker

- Play music for relaxation
- Voice feedback on the daily medication schedule
- Recommendation of daily medicine for older adults
- Physical health alert

Physical Monitor

Detect physical health: blood pressure,
body temperature, heart rate, blood
oxygen, motivation, etc.

Medicine Intake Monitor

Voice Input for Medicine Intake: Tell the box what you eat to track daily medication intake.

Data Upload: Young family members can upload data via the app.

FoodCare System: Use the FoodCare system to collect dietary information.

SeniorCare System Feedback: Provide voice feedback to the **SeniorCare** system to remind older adults to maintain a healthy lifestyle.



Senior-
healthcare



Family members will also receive a message to remote monitor senior's daily meals



<https://images.app.goo.gl/cYbshE25nuEP56xKA>

Other Functions.

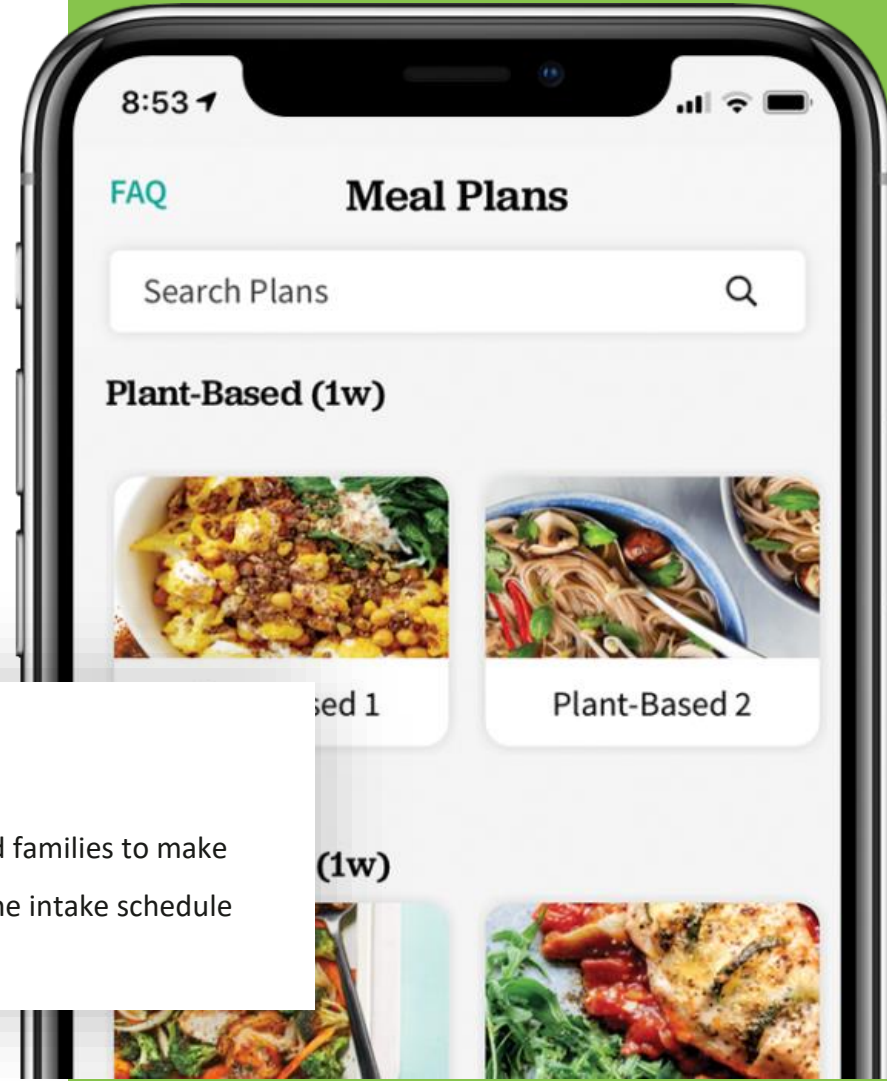
Common medications for older adults include antihypertensives, cholesterol-lowering drugs, antidiabetics, anticoagulants, analgesics, osteoporosis treatments, antidepressants, dementia medications, diuretics, and respiratory drugs, each requiring careful management due to potential interactions and side effects.

01. Guide

diet recommendations according to the medicine category (data) uploaded

02. Plan

helpful for older adults and families to make a plan for the daily medicine intake schedule





Why it is relevant to us.

Family support is important **for older adults**

Older adults often struggle with social isolation, which can significantly impact their mental and physical health. Family support plays a vital role in maintaining connections with loved ones and offering opportunities for social interaction that boost well-being. By actively monitoring health changes and communicating with healthcare providers, family members help ensure older adults receive timely medical attention.



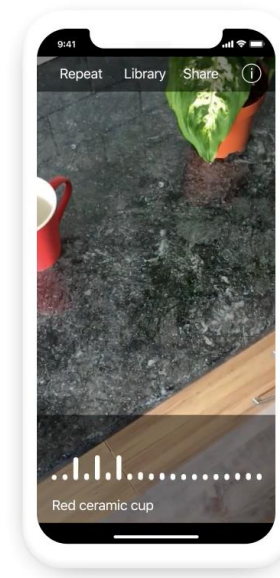
apps
food care

TapTapSee

Assistive Technology for the Blind and Visually Impaired

TapTapSee is a mobile camera application designed specifically for blind and visually impaired users, powered by the [CloudSight Image Recognition API](#). TapTapSee utilizes your device's camera and VoiceOver functions to take a picture or video of anything and identify it out loud for you.

Double-tap the right side of the screen to take a picture or double-tap the left side of the screen to take a video. TapTapSee can accurately analyze and identify any two or three dimensional object at any angle within seconds. The device's VoiceOver then speaks the identification aloud.



Existing Products.

Related to food identification



TapTapSee

Assistive Technology for the Blind and Visually Impaired



SafeFood

combines paperless food safety, supplier & equipment management, automated temperature monitoring and real-time business insight in one convenient dashboard.

- target users: older adults
- function: food identification, classification and selection



■ <http://taptapseeapp.com>
<https://www.safefoodpro.com>



Research Plan.

From June till July

Inquiry

smart home
life experience

Qualitative

Observation

smart home
health care
experience

Qualitative

Survey

food care, medicine
intake monitoring
everyday people

Quantitative

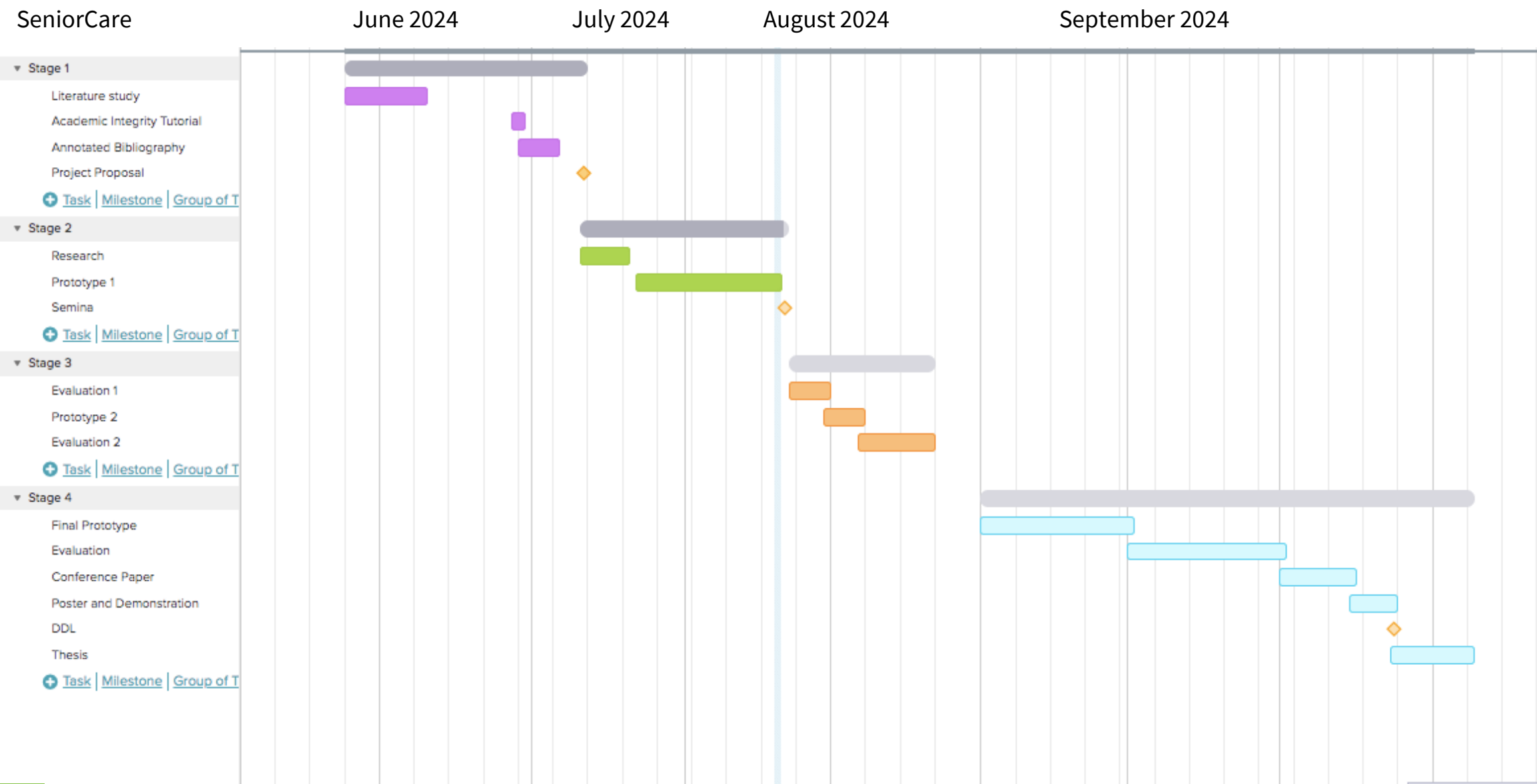
Interview

health care
for older adults

Quantitative

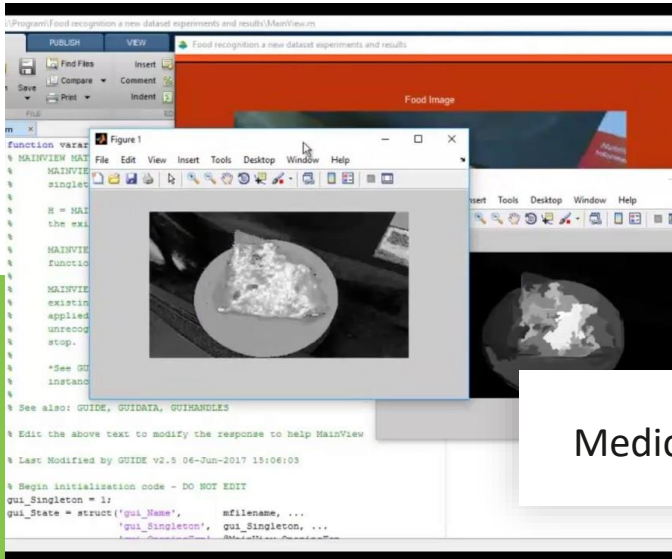
participants aged from 22-35 and above 60, smart technology users (school mates, relatives and friends)

Future Work.



Project **Process.**

Stage	Tasks	Date	Process
Idea generation	Make aspiration matrix, such as mind map Investigate domain problems and theme	22 May-28 May	Done
Paper review	Review 8 literature and write paper review	29 May-4 June	Done
Research	Quantitative and qualitative research including interview and online survey	5 June-18 July	Done
Prototyping and implementation	Use the AI camera and sensors to build a medicine intake monitoring prototype	19 July-16 September	In progress
User evaluation and analysis	Conduct Behavior Design methods and evaluations, including TAM, SUS, interviews, etc. Make graphs and write analysis report	17 September-30 November	To do
Thesis writing	Including draft, image, prototype composing and final copy	31 November-	To do
Submit	Check the submit requirement and final review before due date	31 November-	To do

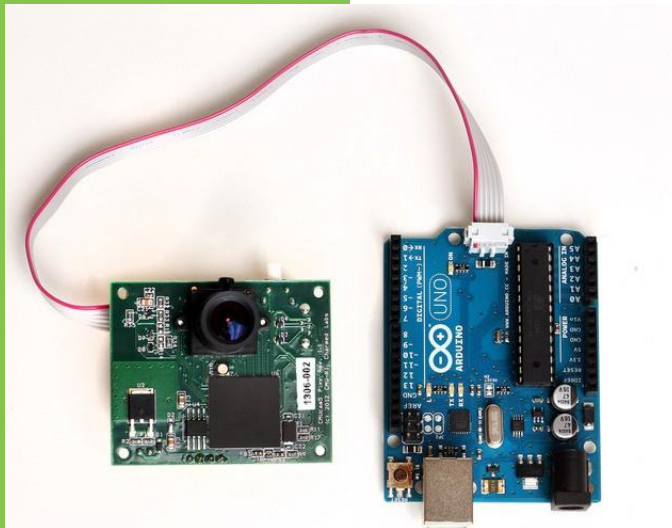
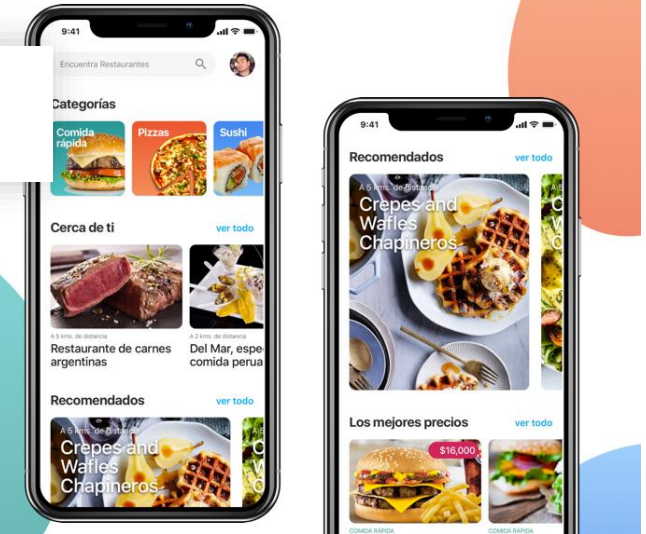


Use machine learning and
Deep Learning Algorithm

Medicine Intake **Monitor**

Interface **Design**

Interface and interaction
for the digital prototype



Other work
need to do.

Use Arduino Uno + Detective sensors
+ Speakers to make a medicine
intake monitoring system

- [1] “How Smart Homes Can Deliver Sustainability as a Service - ProQuest.” https://search-proquest-com.ezproxy.library.uq.edu.au/docview/1873991798?rfr_id=info%3Axri%2Fsid%3Aprimo (accessed Mar. 27, 2020).
- [2] Karolina Gawrońska and Jacek Lorkowski “Smart homes for the older population: particularly important during the COVID-19 outbreak” *Reumatologia*. 2021; 59(1): 41–46. Published online 2021 Feb 28. doi: 10.5114/reum. doi: 2021.103939
- [3] Z. Xu, Y. Dong, and S. Zhu, “Research on System Design of ‘Shared’ Smart Kitchen in Youth Apartment in the Era of Internet,” in *HCI International 2019 - Posters*, Cham, 2019, pp. 536–545, doi: 10.1007/978-3-030-23528-4_73.
- [2] F.-G. Wu and T.-H. Tsai, “Building a Recognition Process of Cooking Actions for Smart Kitchen System,” in *Universal Access in Human-Computer Interaction. Aging and Assistive Environments*, Cham, 2014, pp. 575–586, doi: 10.1007/978-3-319-07446-7_56.
- [3] Z. Xu, Y. Dong, and S. Zhu, “Research on System Design of ‘Shared’ Smart Kitchen in Youth Apartment in the Era of Internet,” in *HCI International 2019 - Posters*, Cham, 2019, pp. 536–545, doi: 10.1007/978-3-030-23528-4_73.
- [5] J. Chuang et al., “SilverLink: Smart Home Health Monitoring for Senior Care,” in *Smart Health*, Cham, 2016, pp. 3–14, doi: 10.1007/978-3-319-29175-8_1.